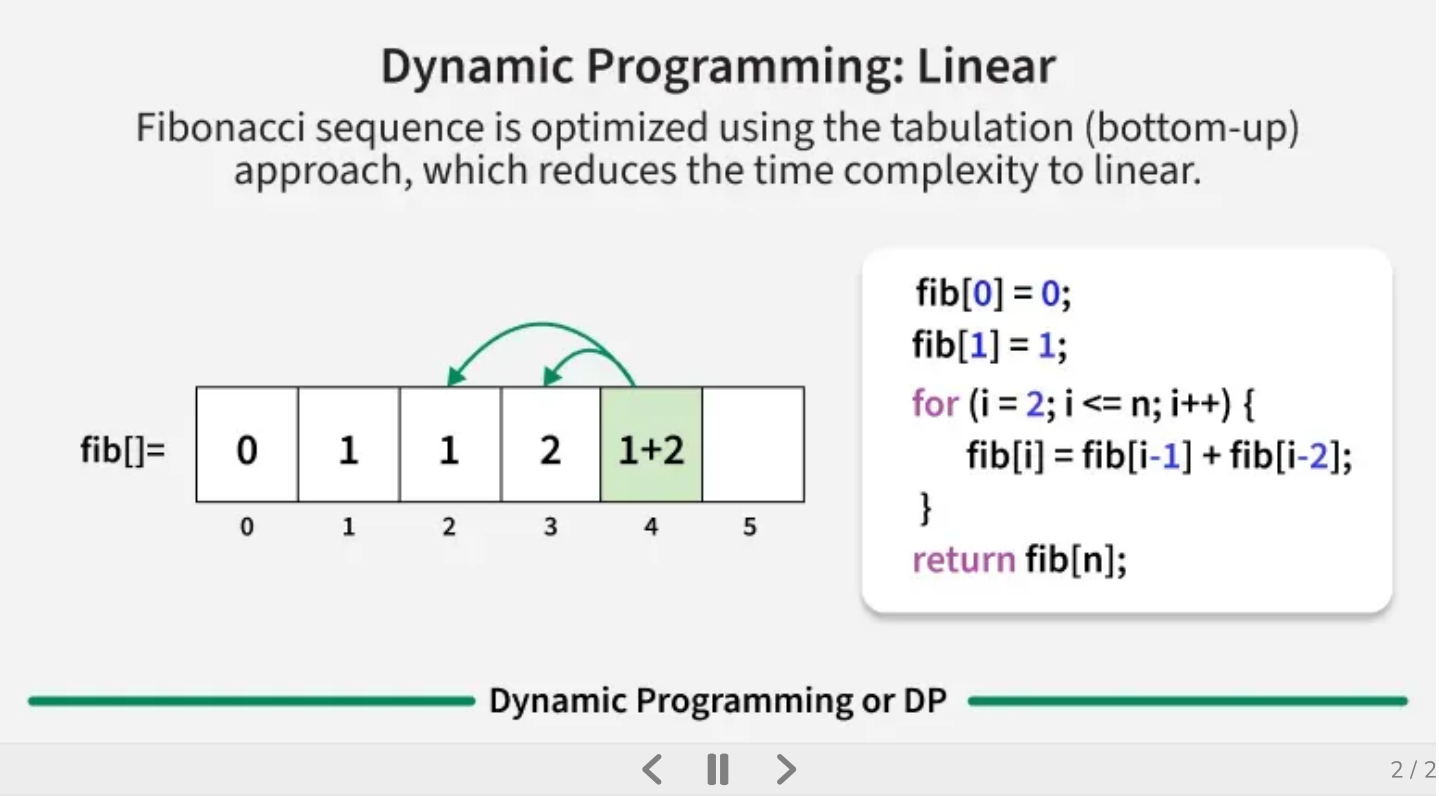
ıic Programming or DP

ed : 26 Jan, 2026

Programming is an algorithmic technique with the following properties.

ainly an optimization over plain recursion. Wherever we see a recursive solution that has repeated call
inputs, we can optimize it using Dynamic Programming.
ea is to simply store the results of subproblems so that we do not have to re-compute them when nee
This simple optimization typically reduces time complexities from exponential to polynomial.
popular problems solved using Dynamic Programming are Fibonacci Numbers, Diff Utility (Longest Co
quence), Bellman–Ford Shortest Path, Floyd Warshall, Edit Distance and Matrix Chain Multiplication.



ıf DP

uction
ation vs Memoization
to solve a DP Problem

Problems

acci numbers
ıacci Numbers
Numbers
ing Stairs
ing Stairs with 3 Moves

ıted Climbing Stairs
ıum Segments
ıtalan Number
Unique BSTs
Valid Parenthesis
to Triangulate a Polygon
ım in a Triangle
um Perfect Squares
to Partition a Set
ıial Coefficient
l's Triangle
ow of Pascal Triangle
ım in a Triangle

roblems

Robber
ost Path
le Ways
t Sum Problem
hange problem - Count Ways
Change – Minimum Coins to Make Sum
ng Fence Algorithm
g a Rod
Game
st Common Substring
all paths in a Grid
in a Grid with Obstacles
ıtations with K Inversions
's using Special Keyboard

n Problems

Overflow
st Common Subsequence
st Increasing Subsequence
istance
st Divisible Subset
ıted Job Schedulling
napsack Problem
ıg Items in 0/1 Knapsack
ınded Knapsack
Break Problem

tacking Problem
tacking Problem
on Problem
st Palindromic Subsequence
st Common Increasing Subsequence (LCS + LIS)
stinct subset (or subsequence) sums
Derangements
um insertions for palindrome
ard Pattern Matching
ar Expression Matching
ge Balls with adjacent of different types
st Subsequence with 1 adjacent difference
num size square sub-matrix with all 1s
an–Ford Algorithm
Warshall Algorithm
num Tip Calculator

Problems

st X Bordered Square
ropping Problem
lrome Partitioning
lromic Substring Count
Wrap Problem
al Strategy for a Game
ainter's partition problem
am for Bridge and Torch problem
Chain Multiplication
g Matrix Chain Multiplication
num sum rectangle
Buy and Sell - At-Most k Times
Buy and Sell - At Most 2 Times
st to sort strings using Reversals
of AP Subsequences
Trees
leight of Tree when any Node can be Root
st repeating and non-overlapping substring
lrome Substrings Count

ms Sorted by Topic / Dimensions

andard Problems and Variations.
oblems Dimension Wise (1D, 2D and 3D)

oblems Topic Wise

ced Concepts

sking and DP1

ing Salesman Problem

DP

ver Subsets

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